Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/20/2006
PI Name:	Boyd, Stephen Ph.D.		
Project Title:	New Phenomena in Strongly Counterflowing He-II near Tlambda		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	87131	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	10/01/1996	**End Date:**	06/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Duncan, Robert (University of New Mexico) Strayer, Donald (Jet Propulsion Laboratory)		
Grant/Contract No.:	None		
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Performance Goal Text:	To Explore the Frontiers of Physics and Unlock Mysteries of the Universe Obscured by Earth’s Gravity.		
	The objective of this research task is to study the thermal behavior of strongly counterflowing helium-II very close to the superfluid transition. An unusual component of this study is the use of a cell which deliberately generates an inhomogeneous heat flow pattern. Our investigation uses a "waisted" experimental geometry which focuses heat flow in a narrow channel to create a volume of thermally resistive helium with unique boundary conditions. The novel experimental geometry also allows the use of thermometers with relatively high bandwidth. We will use this cell to look at two important problems. Firstly, we wish to investigate the anomalous thermal resistance first reported by Liu and Ahlers and recently confirmed by Murphy and Meyer. The anomalous resistance has appeared in only one of the two experimental geometries used so far, thus new measurements performed in the different boundary conditions of the waisted cell are desirable and should give us new insight into this unexplained phenomenon. This study will also extend		

Task Description:	the previous measurements to higher heat fluxes recently explored in a sidewall geometry by Baddar et. al. of Guenter Ahlers' lab. Second, we wish to investigate the onset of dissipation (turbulence) in the superfluid counterflow which transports the heat, under the novel boundary conditions provided by the waisted cell. Third, we wish to investigate time-varying heat flow phenomena in this system, using the relatively high bandwidth of the thermometers. This study will include both passive "listening" and active probing with heat pulses. Our results will be compared with recent predictions of the appearance of nonlinear phenomena, such as thermal solitons, when the counterflow is strong enough to render the superfluid helium thermodynamically unstable. By closely controlling the temperature of the experimental cell close to T-lambda and using a "waisted" cell geometry which focuses a heat flow it is possible to create a helium-II sample containing a resistive volume suspended between two isothermal volumes. Adequate resolution for temperature measurement and thermal control will be realized by using magnetocaloric (paramagnetic salt) thermometry with superconducting quantum interference device (SQUID) readout. The close proximity of the magnetocaloric thermometers requires numerical optimization of the design of the SQUID pick-up coils and the magnetic shields to attenuate magnetic cross-talk between the thermometers. Thermal resistance of the helium is determined by simultaneously measuring the temperatures of the two isothermal volumes for a given heat flux through the cell. Resistive "end effects" due to converging and diverging flow fields at the ends of the narrow channel can be deconvolved by making resistance measurements with at least two different channel lengths. Because the helium-II at the wide ends of the "waisted" cell will be maintained isothermal, immersed magnetocaloric thermometers can be used to measure helium temperatures. Immersed magnetocaloric thermometers have shown outstanding temperature resolution of 5 X 10^-11 K/Hz^.5 near T-lambda and relatively high bandwidth of ~1 kHz. The low noise of the immersed magnetocaloric thermometers requires that special attention be paid to attenuating noise sources throughout construction of the experiment. The relatively high bandwidth allows the cell to be used for measurements of time-varying thermal phenomena. Passive "listening" for time-varying thermal phenomena will be performed during all thermal resistance measurements. Additional measurements will actively probe time-varying thermal behavior with heat pulses. The reversible nature of the magnetocaloric effect may be useful for the generation of probing heat pulses in these additional measurements. This experiment provides unique capabilities in the study of dissipative heat flow in helium-II near the lambda point. Our measurements will further elucidate the complex thermohydrodynamics of liquid helium near the lambda point, and will be a significant contribution to the small fund of high-quality data on critical behavior in highly nonequilibrium systems. Further, vortical superfluid counterflow, the source of thermal resistance in helium-II, is an important limiting case for the general problem of weakly dissipative flow. The role played by coherent vortical structures in weakly dissipative flow is an active area of research. Moreover, the general problem of collective vortex dynamics is also a topic of considerable current interest. The present study of time-varying thermal behavior in resistive helium-II is a timely experiment which may give important insight in these areas. In the context of Microgravity Fundamental Physics, knowledge gained in this work will be relevant to the interpretation of the results from DYNAMX and the bulk-conductivity part of BEST. Knowledge gained about the interaction of second sound with the dissipative heat-flow state may be useful to future microgravity second-sound experiments. In terms of experimental technique, immersed magnetocaloric thermometers have considerable advantages for low-temperature experiments in the space environment. Technical experience gained in this work will provide valuable input for future orbital experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress reported this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)